

Title: UNIVERSAL HEALTH INSURANCE AND CHILD HEALTH: EVIDENCE FROM SEGURO POPULAR

Abstract (150 words)

Government programs designed to increase access to health insurance have rapidly increased in developing countries. This project contributes to filling the gap in what is understood about the consequences of increased access to health insurance in these settings. Specifically, we provide evidence on the persistent impact of Seguro Popular (SP), a universal public health insurance program introduced in Mexico in 2002. By leveraging the temporal and spatial variation created by SP's rollout, and, the biological variation generated by the age of exposure to the program, this project aims to identify the causal effects of SP on child height, a measure consistently associated with indicators of well-being in later life. We find in the first few years of its implementation, SP provided no benefit to child health, but that after 5 years of operation children exposed to SP in early childhood were significantly taller than their unexposed peers.

Extended Abstract

Introduction:

It has been argued that improved access to health coverage yields significant public health and economic benefits. In recent years, there has been a substantial increase in the promotion of universal health insurance (UHI) in developing countries where, relative to advanced countries, access to health services is very limited and the burden of disease is extremely high. Despite these claims, evidence that can be given a causal interpretation and establishes that health insurance access leads to significant improvements in physical health is sparse and almost entirely limited to developed countries. Translating evidence on effects of universal health insurance from developed to developing country contexts is not straightforward. Most relevantly, the type and magnitude of supply (lack of health infrastructure) and demand (poor information distribution, mistrust of healthcare providers) constraints differ significantly.

An additional **critical scientific limitation** is that, with very few exceptions, the literature on access to health insurance in developing countries has only assessed its impact on mortality. Mortality is an extreme and blunt indicator of the effect a program has on population health and likely obscures a substantial portion of the benefits health insurance access provides across the full distribution of health. Moreover, the impact of access to health insurance on population health is likely to accumulate over time, and thus, short-term estimates of the impact on mortality will understate the overall ability of these programs to reduce health disparities and spur economic growth.

The **overarching goal** of our research is to provide new population-representative evidence on the impact of access to health insurance in a developing country context. Our **main objective** is to evaluate the long-term effects of *Seguro Popular* (SP), a national UHI program first implemented in Mexico in 2002, that was designed to cover ~60 million people, almost half the Mexican population, who didn't have insurance coverage. Pairing the innovative design of the Mexican Family Life Survey (MxFLS) with cutting-edge and rigorous non-experimental methods, represents a unique opportunity to make a substantial contribution to our shared understanding of the persistent impact of programs that increase access to health insurance in developing countries.

To isolate the causal impact of access to UHI, we will exploit the fact that the nutrition and biology literature have established that an individual's height is largely determined during their first 1,000 days of life (Martorell and Habicht, 1986). Thus, we will leverage heterogeneity in the timing and location of program implementation, as well as, the child's age of exposure to generate quasi-random in SP exposure intensity. Given the expectation that the introduction of SP increased health care utilization and reduced health care expenditures (Barros, 2008; King et al, 2009, Grogger et al, 2015) and that stature is reactive to changes in nutrition and health resources in childhood, we hypothesize that access to universal healthcare in utero and in early childhood will lead to increased linear growth.

We focus on child height for several reasons. First, it is an important marker of physical health and nutrition in early life that is consistently and widely associated with a broad array of indicators of well-being in adulthood including educational attainment, earnings, health and mortality (Fogel 1994; Strauss and Thomas 1995;

Hoddinott et al. 2008). Second, it is a long-run marker that can be influenced by contemporaneous changes to the health/economic environment. Third, the limited window of ages in which height is sensitive to shocks provides variation that can be leveraged for causal identification. Lastly, the current standard in this literature, mortality, only accounts for the most severe costs of being uninsured and fails to capture any of the other marginal benefits of universal health care. No other health outcome would provide all of these advantages, and thus, we see it as an ideal metric to further gauge the value of UHI programs.

Another important contribution of this research is related to program evaluation over time. Specifically, there are several reasons to believe that the efficacy of a UHI program in a developing country will vary substantially over time as a result of evolving supply and demand side constraints. On the supply side, if the expansion of the population covered by insurance in a community precedes increases in health infrastructure, services, and providers to meet elevated demand in the community, immediate and longer-term effects are likely to differ. On the demand side, it may take time for newly covered individuals to use public health services because of poor information about who is covered, what is covered, or the benefits of coverage. Therefore, we will utilize the three waves of the Mexican Family Life Survey (MxFLS), a large-scale nationally representative longitudinal study that spans the temporal and spatial rollout of SP to measure the impact of the program over time. We hypothesize that the positive impact of SP will develop as the program ages and these constraints diminish. In addition, we hypothesize that the speed with which a community experiences the full benefit of the program will be positively related to the size of its pre-program healthcare infrastructure and the percentage of the population that was already utilizing the healthcare system.

Literature Review

Rigorous examination of the health benefits of access to health insurance in developing countries is limited to its impact on mortality (Aquino, de Oliveira, and Barreto, 2009; Gruber, Hendren, and Townsend, 2014; Cesur et al, 2017; Bhalotra, Rocha, and Soares, 2019; Conti and Ginja 2023). Whereas mortality is an important outcome, it provides narrow insight into the relationship between health insurance and the health status of the population for at least three reasons. First, mortality, a binary indicator, is a blunt measure of health that is likely relevant for only the most vulnerable, and thus does not capture fine-grained health effects across the entire distribution of health. Second, since mortality is a low-frequency event, its use as a measure of population health may lack power to detect significant effects. Third, the effects of access to health insurance on population health likely accumulate over time and short-term estimates of the impact on mortality will understate the overall benefits of UHI to health disparities and economic growth.

These concerns apply both broadly to the development literature on universal health care as well as specifically to research on Seguro Popular (SP). For example, an experimental evaluation of SP used a matched-cluster design conducted in a small number of carefully selected municipalities and concluded that the program resulted in reduced healthcare expenditure but found no effects on health outcomes, health care utilization, or medication spending (King et al, 2009). Beyond concerns about the representativeness of the selected communities, this study was designed to provide evidence on only the short-term effects of the program, because within 10 months of being introduced in treatment areas, the program was also implemented in control clusters (King et al, 2007). This highlights a primary concern with experimental studies that leverage the phased introduction of a program or policy: **it is not possible to identify long-term effects or measure how effects evolve over time without making additional assumptions that are typically difficult to test.**

Observational studies of SP have compared health outcomes before and after its introduction in 2002. Some studies report positive health benefits (Sosa-Rubí et al. 2009b; Pfutze 2014; Pfutze 2015; Bleich et al. 2007, Knox 2016) while others report no impact (Barros 2008, Turrini et al. 2016). In addition to these results being mixed, the estimates cannot be given a causal interpretation because they rely on the assumption that the timing of the SP rollout was random. The validity of that assumption has been called into question as the program was designed to prioritize states with greater need for health services (Conti and Ginja 2023).

Recent work by Conti and Ginja (2023) provides the strongest evidence to date on the impact of SP on mortality. They find that, overall, there is no impact on mortality. However, in poor municipalities, SP was associated with a 10% reduction in infant mortality and a 4% decrease in mortality among individuals aged 20-59. Importantly, these benefits, became evident only three or more years after program implementation, highlighting the critical importance of measuring the effects of the program over the medium and longer term.

Background Information on Seguro Popular:

In 2002, the Mexican government introduced SP, to address large and long-standing inequalities in health care access and health outcomes. The program intended to cover the almost 60 million people that did not have insurance coverage. SP was designed to provide access to broad set of preventive and curative health services, treatments and, medications with estimated cost of ~1% of GDP

Prior to SP, the vast majority of Mexican families covered by health insurance were employed in the formal or public sector, with ~40% of Mexican citizens covered by IMSS (formal sector insurance), 7% by ISSSTE (public sector insurance), and 3% by private insurance. Healthcare was also available to the poor through two programs. PAC, started in 1996 and consisted of health brigades visiting the most rural & marginalized areas of the country and Progrespa, launched in 1997 in rural areas was an antipoverty CCT program in Mexico & was renamed Oportunidades when it expanded to urban areas in 2002.

Despite these programs almost half of the Mexican population did not have insurance coverage. While the uninsured not covered by PAC or Progrespa could seek healthcare, payment was at the point of use and patients had to buy their own medications, making utilizing healthcare resources very expensive.

Initial piloting and implementation occurred in 20 (of 32) states in the 4th quarter of 2002 (the other 12 initiated between 2003-2005). States w/ more uninsured households, a higher poverty rate, greater enrollment demand, as well as, a sufficient budget & the ability to provide services were to be targeted first. Within states, SP was rolled out at the municipality level based on similar criteria.

SP was originally designed with a few core features. First, upon enrollment, households would be assigned to a PCP and a public health clinic. Second, every enrollee would have access to services without a co-payment at the point of care. Third, SP would provide a wide catalogue of medications, including ~400 drugs by 2012 and lastly, a government managed fund would be available to cover catastrophic events, including 50+ interventions such as cancer treatment or surgery.

Identification Strategy for the Effect of SP in its First 3-4 Years:

To credibly isolate the causal impact of access to UHI, we exploit the following facts: 1) the nutrition and biology literature have established that an individual's adult height is largely determined by age 3 (Martorell and Habicht 1986; Frankenberg et al. 2005); 2) the baseline wave of the MxFLS, MxFLS1, was collected in early 2002 before any respondent had access to SP; 3) the first and largest wave of SP implementation occurred in 20 of 32 Mexican states in mid to late 2002; 4) The first MxFLS survey follow-up, MxFLS2, occurred at a time, 2005-2006, in which there were still many municipalities that had not yet implemented SP.

Thus, one estimate of the impact of exposure to SP on child height, given age and gender, could compare height in MxFLS2 (approximately three years after SP's implementation in 2002) of children who were exposed to the program during their first 1,000 days of life with the height of children from the same municipalities who were too old for their linear growth trajectories to be affected by exposure to the program. However, this estimate will be biased if there are differences in the relative height of children across cohorts. To address this concern, we can compare this estimated difference in height with an analogous difference in height of the same age cohorts in the same communities but measured using MxFLS1, before SP had been introduced. This second difference captures any fixed cohort difference in height for age within these communities. Empirically, we estimate the following model:

$$hfa_{is} = \beta_0 + \beta_1 age0 - 2_i + \beta_2 MxFLS2_s + \beta_3 (age0 - 2 * MxFLS2)_{is} + u_{is} \quad (1)$$

where hfa_{is} represents the height for age z-score, for individual, i , measured in survey year, s , $age0-2_i$ represents an indicator for whether the individual is 0 to 2 years old at the time of the survey, and $MxFLS2_s$ represents an indicator for whether the measurement took place in the MxFLS2 survey round.

By using location of residence information from MxFLS1, this model will be restricted to children whose parents, prior to SP implementation, were already living in a municipality that would later adopt SP in 2002. By assigning treatment exposure/timing based on where the individual's parents were living in MxFLS1, before the program's introduction, we estimate an intent-to-treat effect that is unaffected by potentially bias-inducing selective migration. In addition, we limit the sample to compare children whose height has the most potential to benefit from SP, 0-2 years old in MxFLS2 from municipalities that introduced SP in 2002 (i.e. fully exposed to SP

throughout the critical 1,000 day window) with children aged 10 to 12 years old in MxFLS2 who were not exposed to SP at any point during their first 1,000 days of life. In (1), the DiD estimate β_3 represents: (height of 0-2 year olds - height 10-12 years olds)_{MxFLS2} - (height of 0-2 year olds - height 10-12 years olds)_{MxFLS1}. The first difference combines the average effect of treatment and any fixed cohort differences, while the second difference represents only the fixed cohort differences, and thus, under the parallel trend assumption, the DiD isolates the effect of treatment. Lastly, standard errors will be clustered at the rollout group, age, and survey wave level as that is the level of treatment assignment.

This DiD approach is not only useful in its own right, but it also provides a framework to test the potential limitations of its assumptions. For example, if there is concern that cohort height differences or how height was measured evolved over time differently by age group, which would bias the DiD approach, we can add in a third difference. The third difference exploits the same cohort comparisons as in (1), but in communities where SP had not yet been introduced by MxFLS2. That triple difference is estimated using the following model:

$$hfa_{ism} = \delta_0 + \delta_1 age0 - 2_i + \delta_2 MxFLS2_s + \delta_3 SP2002_m + \delta_4 age0 - 2 * MxFLS2_{is} + \delta_5 (age0 - 2 * SP2002)_{im} + \delta_6 (MxFLS2 * SP2002)_{sm} + \delta_7 (age0 - 2 * MxFLS2 * SP2002)_{ism} + u_{ism} \quad (2)$$

where hfa_{ism} represents the height for age z-score, for individual, i , measured in survey year, s , living in a municipality m , that implemented SP in 2002. Therefore, $SP2002_m$ represents an indicator for whether the individual's parents were living at baseline in a municipality that would later implement SP in 2002. The sample now includes individuals whose parents either lived in a municipality that adopted SP in 2002 or a municipality that does not adopt until after the MxFLS2 survey wave.

The triple difference estimator is represented by δ_7 . The inclusion of these additional comparison groups will provide an assessment of model (1)'s robustness to concerns about the parallel trends assumption. In addition, we can also conduct a test of our identifying assumptions by estimating a placebo test version of models (1) or (2) that only utilizes individuals that were too old to have had their height impacted by SP. Most flexibly, to ensure our results are not driven by more complex bias-inducing pre-trends, or by the specific use of 10-12 year olds as the control group, we will also generate event study versions of models (1) and (2) in which we lift the age cohort sample restriction and estimate:

$$hfa_{is} = \gamma_0 + \sum_{j \neq 6} \varphi_j I(age = j)_i + \gamma_1 MxFLS2_s + \sum_{j \neq 6} \theta_j I(age = j) * MxFLS2_{is} + u_{is} \quad (3)$$

$$hfa_{ism} = \gamma_0 + \sum_{j \neq 6} \varphi_j I(age = j)_i + \gamma_1 MxFLS2_s + \gamma_2 SP2002_m + \sum_{j \neq 6} \tau_j I(age = j) * MxFLS2_{is} + \sum_{j \neq 6} \eta_j I(age = j) * SP2002_{im} + \gamma_3 MxFLS2 * SP2002_{sm} + \sum_{j \neq 6} \theta_j I(age = j) * MxFLS2 * SP2002_{ism} + u_{ism} \quad (4)$$

where $I(age=j)$ represents a set of age at survey indicators. The excluded age cohort, age 6, serves as the comparison cohort to test the parallel trends assumption and explore treatment profile dynamics.

Importantly, in every model, only one treatment timing event is used for variation, SP implementation in 2002, and thus our estimates are not at risk of the bias that has been documented in staggered DiD designs (Goodman-Bacon 2021). This emerging literature is only relevant when an analysis incorrectly uses earlier treated cohorts as a "control" group for individuals exposed to a later wave of treatment.

Two final potential threats to identification merit discussion. First, our analysis would not be appropriate for our research question if other government programs that could impact child health overlapped with SP in terms of timing and rollout. While there are no direct threats that exist on all of these dimensions, the urban component of the large social assistance programs, Oportunidades (OP), was founded in 2002. While, it has been shown that the urban phase of OP had no impact on child height (Farfan Bertran et al. 2022), in order to confirm that our analyses are not contaminated by the rollout of OP in urban areas we will check that our conclusions are robust to restricting to rural areas in which OP was implemented 5 years prior to SP and stratifying the sample based on pre-SP enrollment in OP.

Second, our estimates could be contaminated by non-random composition differences in our treatment and control groups. Despite the quasi-random variation we are exploiting and the identification strengths of DiD, triple difference, and event study designs there are three potential ways selectivity into the sample could cause bias: attrition, mortality, and fertility. The MxFLS is particularly robust to issue of attrition as retention rates throughout the three waves of the survey are above 90%. In addition, we will be able to directly test if the attrition rate is correlated with the interaction of SP exposure, survey wave, and age at survey.

Mortality selection is a potential concern since Conti and Ginja (2023) document that the introduction of SP led to a reduction in infant mortality. This type of selection would bias our results against finding a positive impact of SP on child health outcomes since the additional children in the treatment group would be drawn from the lower end of the in-utero health distribution: we will, therefore, examine the number of live births, birthweight, and family size of the treatment group directly. Similarly, with improved access to healthcare, more disposable income due to reductions in health expenditures, and less infant mortality, families may alter their fertility decisions as a result of SP. To explore potential bias from these sources of non-random sample selection, we will replace child height with parental characteristics as the outcomes in our models. Preliminary results from models (1) and (2) show that there are no statistically significant differences between our treatment and control children on over 35 parental and household characteristics measured during the baseline survey.

Identification Strategy for the Effect of SP More Than 5 Years After Implementation:

It has been suggested that the impact of a UHI program, like SP, varies over time because of supply or demand constraints that exist when the program is implemented (Atun et al, 2015). Our hypothesis is that the positive impact of SP grows as the program ages and that a community's benefit from the program is positively related to its pre-program healthcare infrastructure and healthcare utilization rate.

In order to provide novel insight into the way the impact of a large public program in a developing country, like SP, may evolve as it ages, we leverage the fact that the MxFLS's second and third survey waves (2005-2006 and 2009-2010, respectively) can provide a window into how the program's effects have changed after three years versus six (or more) years of existence. The MxFLS allows us to be uniquely positioned to observe the effect of SP using the same municipalities, same age groups, same identification strategy, and sometimes, even the same children to examine this fundamental question about the efficacy of UHI in a developing country. To do this we will replicate models (1) and (3) using MxFLS1 and MxFLS3 to estimate:

$$hfa_{is} = \beta_0 + \beta_1 age0 - 2_i + \beta_2 MxFLS3_s + \beta_3 age0 - 2 * MxFLS3_{is} + u_{is} \quad (5)$$

$$hfa_{is} = \gamma_0 + \sum_{j \neq 10} \varphi_j I(age = j)_i + \gamma_1 MxFLS3_s + \sum_{j \neq 10} \theta_j I(age = j) * MxFLS3_{is} + u_{is} \quad (6)$$

where MxFLS3_s represents an indicator for whether the measurement took place in the MxFLS3 survey round.

The estimates of models (5) and (6) relative to (1) and (3) will provide us a rare opportunity in an impact evaluation setting to assess the efficacy of a public program at varying stages of its existence, which as detailed previously, is likely to be highly relevant for UHI in this context. While models analogous to (2) and (4) cannot be estimated using the MxFLS3 survey, as all 0-2 year olds will have been exposed to SP by 2009, model (6) will offer a strong indication of whether the parallel trends assumption holds in this case. Additionally, comparing models (1) and (2) can provide a guide for the size and direction of any bias the triple difference model would have eliminated.

We will complement these analyses by exploiting the MxFLS's community and individual surveys. These data will allow us to explicitly explore the relative importance of demand and supply side constraints by conducting heterogeneity analysis of models (1)-(6) using pre-SP health infrastructure and self-reported healthcare utilization.

Preliminary Results:

Estimates of the base models (1) and (2) as well as extended models that include household/parental controls and temporal and geographic fixed effects can be found in Tables 1 and 2. In both cases, SP appears to have no significant impact on child height when evaluated 3-4 years after the program began. In addition, Figure 1, which estimates the event study design from equation (4) provides strong evidence that our findings are not driven by our specific choice of control cohorts or by underlying trends in child health in treatment communities.

When evaluating the impact of exposure to SP in early childhood when SP has been operating for more than 5 years, we estimate equation (5) whose results can be found in Table 3. These findings present a very different story regarding the benefits of SP for child health. When evaluated more than 5 years after implementation, SP exposure in early childhood is associated with a significant and non-trivial increase in the height-for-age of treated kids vs. untreated kids in MxFLS3. Specifically, exposure to SP from 0-2 years old is associated with a .2-.38 standard deviation increase in height-for-age. Examination of the event study design estimates of equation (6) in Figure 2 confirm these results are again not the product of the choice of comparison cohort or by underlying trends in child health in treatment communities.

Next Steps:

As we continue to work on this project we plan to:

- 1) Explore heterogeneity in the estimated treatment effects related to the child's parental characteristics measured at baseline including parental employment, parental insurance coverage, parental income, parental education, as well as, characteristics that describe the health and healthcare characteristics of their community.
- 2) Examine behavioral responses to SP such as fertility, migration, labor market attachment, employment sector choice, insurance take-up, healthcare utilization, healthcare expenditure, and health behaviors.
- 3) Try to measure and investigate the role of supply and demand frictions at the early and later implementation period of SP. To assist with this we can use survey data on the availability, quality, & prices of health care in each municipality.

Table 1
D-i-D of Full Exposure to SP During its First 3-4 Years
Roll Out: 4Q2002; Age Diff: (0-2)-(10-12); Survey Diff: Mx2 - Mx1

	[1]	[2]	[3]	[4]	[5]
All children	-0.049 (0.170)	0.032 (0.159)	0.033 (0.149)	0.056 (0.146)	0.029 (0.140)
Observations	1,668	1,668	1,668	1,668	1,668
Mean of DV	-0.32	-0.32	-0.32	-0.32	-0.32
Child & HH Chars	NO	YES	YES	YES	YES
Municipality FEs	NO	NO	YES	YES	YES
Interview Date FEs	NO	NO	NO	YES	YES
Age FEs	NO	NO	NO	NO	YES

Notes: Robust variance-covariances clustered at roll out group, age and survey wave level.

Child controls include gender and rural status and household controls include mother and father's baseline household size, household pce, rural status, height, age, cognition, health status, marital status, years of education, employment status, earnings last month, weight, formal employment status, and access to insurance.

Table 2

Triple Difference of Full Exposure to SP During its First 3-4 Years
Roll Out: 4Q2002 – 2005+; Age Diff: (0-2)-(10-12); Survey Diff: Mx2 - Mx1

	[1]	[2]	[3]	[4]	[5]
All children	0.002 (0.292)	0.112 (0.238)	0.113 (0.211)	0.111 (0.212)	0.099 (0.212)
Observations	3,334	3,334	3,334	3,334	3,334
Mean of DV	-0.57	-0.57	-0.57	-0.57	-0.57
Child & HH Chars	NO	YES	YES	YES	YES
Municipality FEs	NO	NO	YES	YES	YES
Interview Date FEs	NO	NO	NO	YES	YES
Age FEs	NO	NO	NO	NO	YES

Notes: Robust variance-covariances clustered at roll out group, age and survey wave level.

Child controls include gender and rural status and household controls include mother and father's baseline household size, household pce, rural status, height, age, cognition, health status, marital status, years of education, employment status, earnings last month, weight, formal employment status, and access to insurance.

Table 3
D-i-D of Full Exposure to SP 5+ Years After Implementation
Roll Out: 4Q2002; Age Diff: (0-2)-(10-12); Survey Diff: Mx3 - Mx1

	[1]	[2]	[3]	[4]	[5]
All children	0.374*** (0.132)	0.305** (0.138)	0.303** (0.124)	0.279** (0.115)	0.206* (0.110)
Observations	1,762	1,762	1,762	1,756	1,756
Mean of DV	-0.19	-0.19	-0.19	-0.19	-0.19
Child & HH Chars	NO	YES	YES	YES	YES
Municipality FEs	NO	NO	YES	YES	YES
Interview Date FEs	NO	NO	NO	YES	YES
Age FEs	NO	NO	NO	NO	YES

Notes: Robust variance-covariances clustered at roll out group, age and survey wave level.

Child controls include gender and rural status and household controls include mother and father's baseline household size, household pce, rural status, height, age, cognition, health status, marital status, years of education, employment status, earnings last month, weight, formal employment status, and access to insurance.

Figure 1

**Event Study Triple D of Full Exposure to SP During its First 3-4 Years
Roll Out: 4Q2002 – 2005+; Age 6 Ref. Cohort; Survey Diff: Mx2 - Mx1**

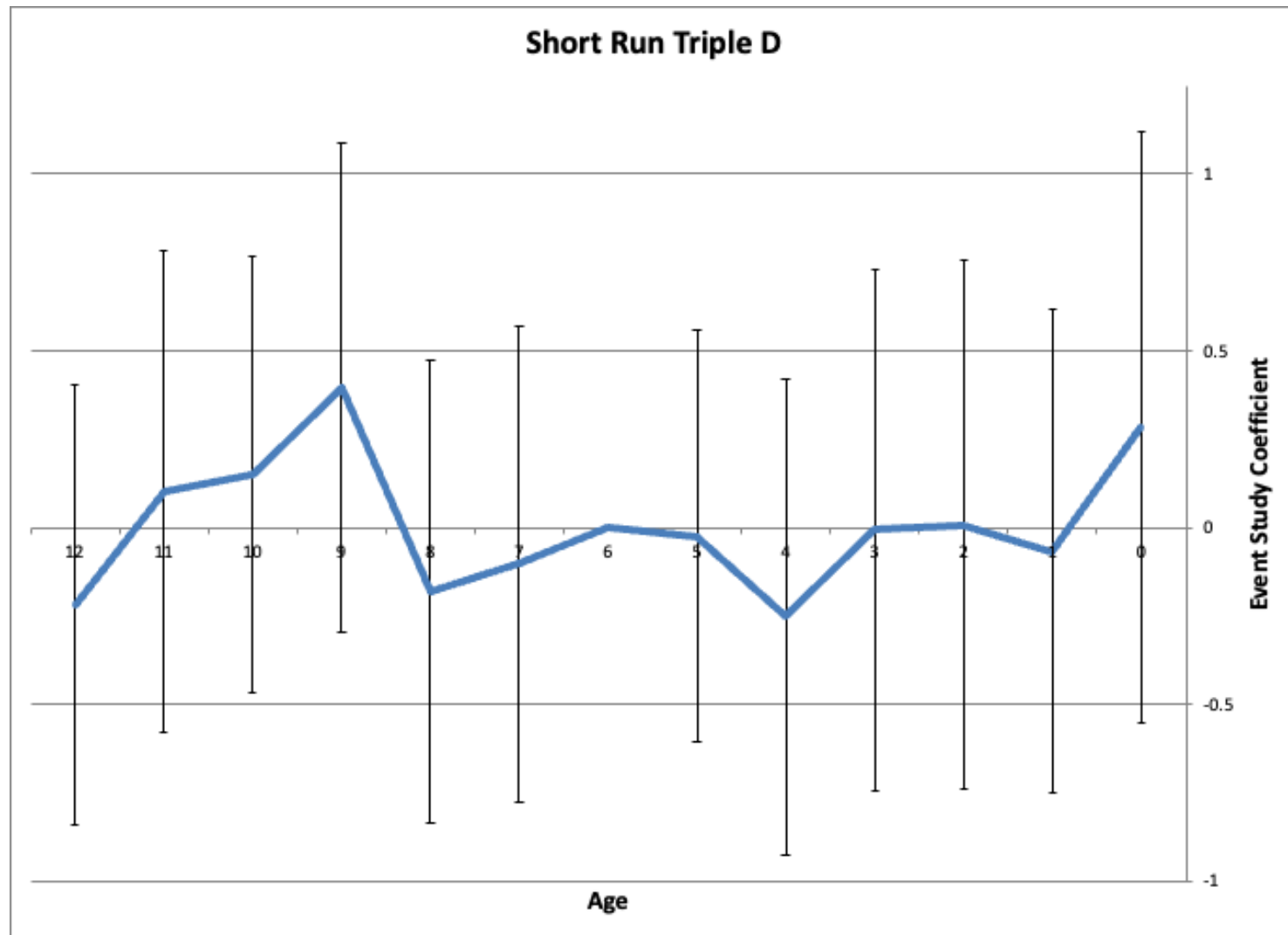


Figure 2

**Event Study DiD of Full Exposure to SP 5+ Years After Implementation
Roll Out: 4Q2002; Age 10 Ref. Cohort; Survey Diff: Mx3 - Mx1**

